

PIERRE-LOUIS SIXDENIER

Embedded Systems Engineer • PhD Candidate

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EXPERIENCE

[Job Title Placeholder • e.g. PhD Researcher / R&D Engineer]

November 2020 – May 2026

[Company / Research Institution / University Lab Placeholder]

France

- [Placeholder for research activities, engineering deliverables, or doctoral thesis work between Nov 2020 – May 2026].

FPGA R&D Engineer Intern

May 2020 – Nov 2020

Safran Electronics & Defense

France

- Conception of a debugging streaming IP for an embedded low-power FPGA platform.

Research Intern

December 2018

ETIS Laboratory (Equipes Traitement de l'Information et Systèmes)

Cergy, France

- Conception and implementation of a failure detection system for robots.

Research Intern

June 2017

XLIM-SIC Laboratory

Poitiers, France

- Conception of a proof-of-concept (PoC) on an L-System generator.

EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science

2020 – 2026 (Exp.)

[Doctoral School / Research Lab Placeholder]

France

- Research in Embedded Systems, Hardware Acceleration & Intelligent Sensor Computing.

Master in Computer Science

2018 – 2020

CY Cergy Paris University

Cergy, France

- **Top-ranking of the research program in Smart Electronic Systems.** Advanced curriculum in FPGA, VHDL, and embedded systems.

Exchange Program • Computer Science

2017 – 2018

Oregon State University

Corvallis, OR, USA

- Followed CS classes and gave French lessons to undergraduate students.

Licence in Computer Science

2015 – 2018

University of Poitiers

Poitiers, France

- Graduated with high honors.

PROJECTS

Personal Projects:

- **Smart glasses:** An HUD on an OLED screen mounted on glasses communicating with an Android device through BLE for navigation assistance.
- **Pollution advisor:** An Android app and a small Arduino device with sensors to crowdsource an open database about pollution of cities.
- **Foot angle detector:** An Android app that computes the angle of someone's foot on an ArUco board (therapeutic use).

Academic Projects:

- **Spiking Neural Network on FPGA:** Research and simulation of a Spiking Neural Network (SNN) on an FPGA platform.
- **“La futée” – Smart Car:** A driving assistance system that anticipates dangers ahead through sensors and warns the user.

SKILLS & RECOGNITIONS

Technical Languages: C++, C, VHDL, Python, Node.js, HTML/CSS

DBMS: MongoDB, Neo4J, PostgreSQL, MySQL

Embedded & FPGA: Arduino, STM32 (Keil µVision), Vivado, Quartus

Programming Flow: Git, Docker, UML

Spoken Languages: French (Native), English (C1 / IELTS: 7 / TOEIC: 975), Spanish (B1)

Proud of: 2nd place at Renault Digital Hackathon 2019 • 2nd place at ENSEack 2020